EXPERIMENT 5

Interactive Dashboard Development using Tableau / Microsoft Power BI

Answers only — with illustrative charts and dashboard output mockup

1. What is Business Intelligence (BI)? How does it support organizational decision-making?

Business Intelligence (BI) is the process of collecting, organizing, analyzing, and presenting data to turn it into useful information. BI tools help organizations monitor performance, identify patterns, compare results, and make evidence-based decisions. For example, the employee dataset used in Experiment 4 can help a manager compare department sizes and examine employee attrition rather than relying on assumptions.

2. What is an interactive dashboard? How is it different from a static report?

An interactive dashboard is a visual display of important measures and charts that users can explore using filters, slicers, and drill-down controls. A static report shows fixed results, while an interactive dashboard lets users change the view and investigate details without creating a new report each time.

3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Aspect	Tableau	Microsoft Power BI
Visualization	Strong, flexible visual exploration and chart customization	Broad visual options with close integration to Microsoft products
Usability	Visual, drag-and-drop workflow; can take practice for advanced features	Familiar for many Excel users; data modeling uses Power Query and DAX
Sharing / ecosystem	Tableau Cloud/Server and Tableau Public options	Power BI Service, Teams, Excel, and Microsoft 365 integration
Typical uses	Exploratory analysis, visual storytelling, enterprise dashboards	Operational reporting, KPI monitoring, Microsoft-centered business reporting

Both tools support data connection, KPIs, interactive visuals, and dashboard sharing. The best choice depends on the organization’s existing systems, licensing, user skills, and reporting needs.

4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

Key Performance Indicators (KPIs) are measurable values used to evaluate progress toward a goal. Sales KPIs include total sales revenue, total profit, number of orders, average order value, sales growth, conversion rate, and customer count. For the HR dataset in Experiment 4, equivalent dashboard KPI cards can show total employees, attrition rate, average age, and average daily rate.

5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

Filters restrict the data shown to selected values. Slicers provide a visible, user-friendly control for making those selections. Drill-down lets a user move from a summary to more detailed levels, such as from all employees to a department and then to an age group. These features help users answer follow-up questions while keeping the same dashboard.

6. What factors should be considered while designing an effective business dashboard?

A good dashboard should have a clear purpose, relevant KPIs, accurate and refreshed data, readable labels, appropriate chart types, consistent formatting, logical layout, accessible colors, useful filters, and limited clutter. The most important measures should be visible first, and every chart should help answer a business question.

7. How can dashboards help managers monitor organizational performance in real time?

When connected to frequently refreshed data, dashboards can display updated KPIs and highlight changes or exceptions. Managers can compare performance by team, product, region, or time period and investigate unusual results quickly. Actual real-time monitoring depends on the data connection and refresh settings; not every dashboard updates instantly.

8. Why is data visualization important in Business Intelligence tools?

Data visualization presents information through charts, graphs, maps, and KPI cards. It makes trends, differences, relationships, and outliers easier to notice than raw tables alone. Appropriate visualizations also help communicate findings to people who may not work directly with the underlying data.

9. What insights can be derived from a sales dashboard for an e-commerce company?

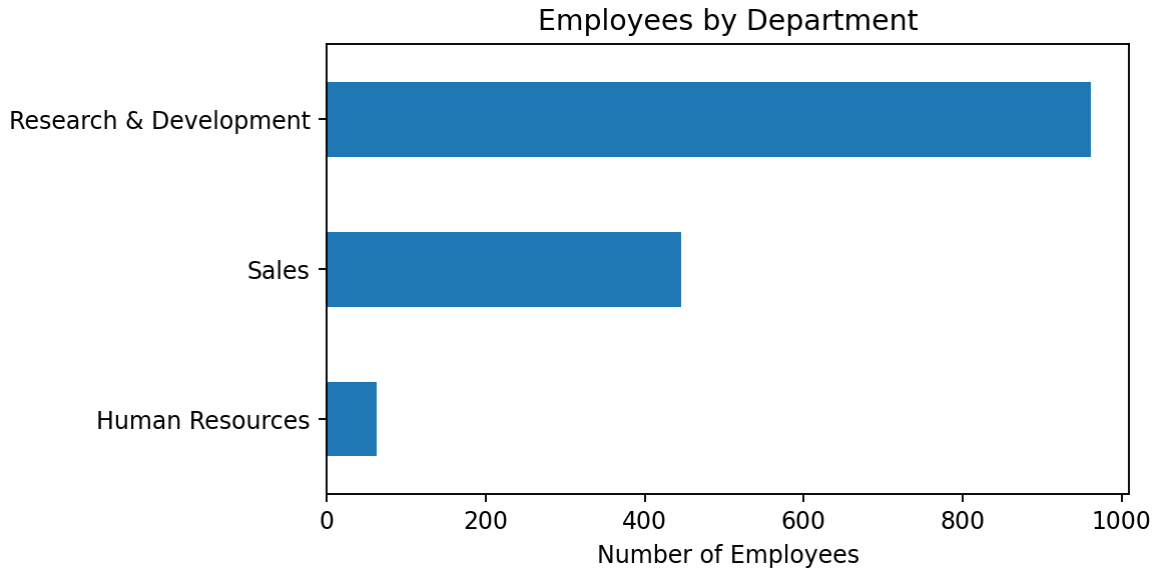
A sales dashboard can identify best-selling products, high-revenue regions, profitable customer segments, changes in sales over time, average order value, repeat-purchase patterns, and products with weak performance. Combining sales and profit measures is important because high revenue does not always mean high profit. The exact findings depend on the company’s dataset.

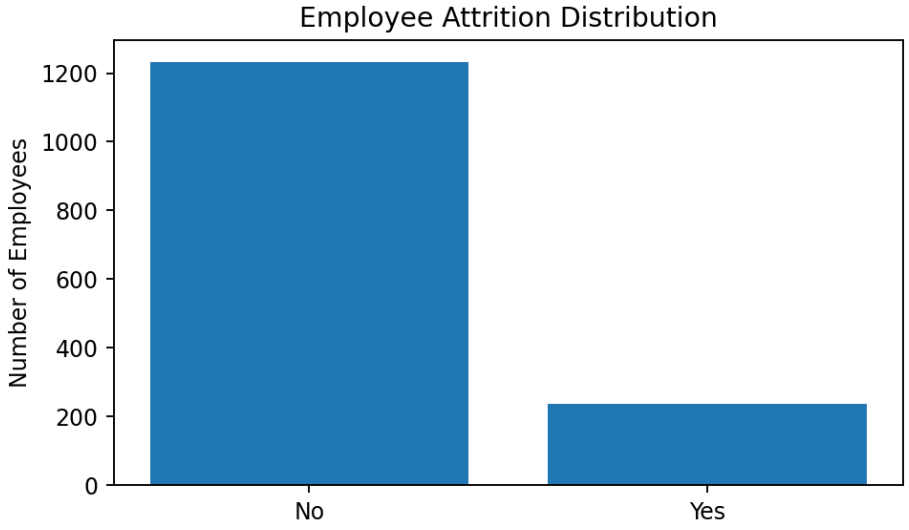
10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Interactive dashboards give analysts and stakeholders a shared, visual view of the same metrics. Stakeholders can explore filters and ask more specific questions during discussions, while analysts can explain trends using consistent definitions and data. This reduces reliance on repeated static reports and supports clearer, faster conversations.

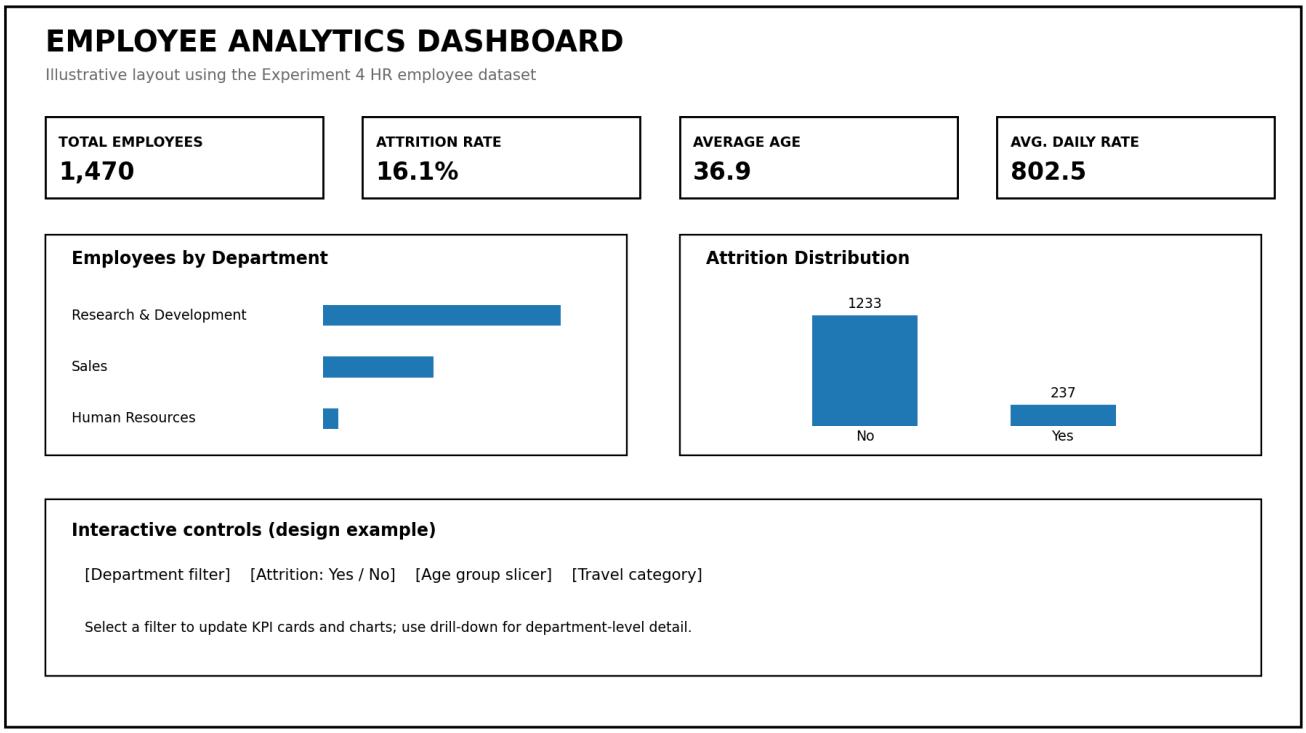
ILLUSTRATIVE CHARTS AND OUTPUT

The following visuals use the employee attrition summary shown in Experiment 4 as the reference dataset. They are illustrative dashboard outputs, not screenshots captured from Tableau or Power BI.





DASHBOARD MOCKUP



Suggested workflow



Example KPI calculation: Attrition rate = employees marked “Yes” ÷ total employees × 100 = 237 ÷ 1,470 × 100 ≈ 16.1%. Average age ≈ 36.9 years. Department counts: Research & Development 961, Sales 446, and Human Resources 63.

Note: The question sheet describes dashboard development with sales-oriented sample datasets, while the supplied Experiment 4 reference uses an HR employee attrition dataset. Therefore, the charts and mockup demonstrate dashboard structure using the available HR data; sales-specific insights are answered conceptually.